Quadratic Kernel Mapping and Kernel Trick

$$\Phi(x) = [2x_1x_2 \;\; x_1^2 \;\; x_2^2]$$ where: $\Phi(x) = [2x_1x_2 \;\; x_1^2 \;\; x_2^2]$

$$\Phi(a) \cdot \Phi(b) = [2a_1a_2 \;\; a_1^2 \;\; a_2^2] \cdot [2b_1b_2 \;\; b_1^2 \;\; b_2^2] = 2a_1a_2b_1b_2 + a_1^2b_1^2 + a_2^2b_2^2 = (a_1b_1 + a_2b_2)^2 = (a \cdot b)^2$$

$$K(a, b) = (a \cdot b)^2$$

$$(a \cdot b)^2 = (a_1b_1 + a_2b_2)^2 = a_1^2b_1^2 + 2a_1b_1a_2b_2 + a_2^2b_2^2$$

$$= [a_1^2 \;\; \sqrt{2}a_1a_2 \;\; a_2^2] \cdot [b_1^2 \;\; \sqrt{2}b_1b_2 \;\; b_2^2]$$

$$= \Phi(a) \cdot \Phi(b)$$

$$K(a, b) = (a \cdot b)^2$$

$$\Phi(a) \cdot \Phi(b) = (a \cdot b)^2 = K(a, b)$$